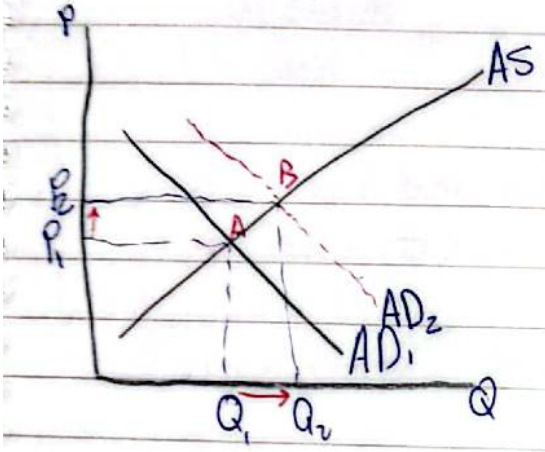
$\uparrow AD = C + I + G \uparrow$

- elasticity of supply
- kind of expenditure
  - current exp
  - Capital exp

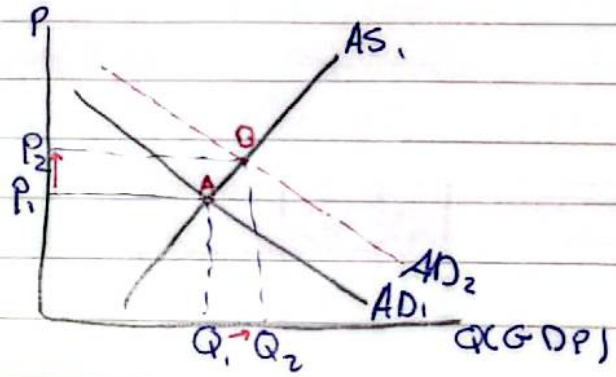
• Current exp: $\uparrow G$

A-elastic supply curve



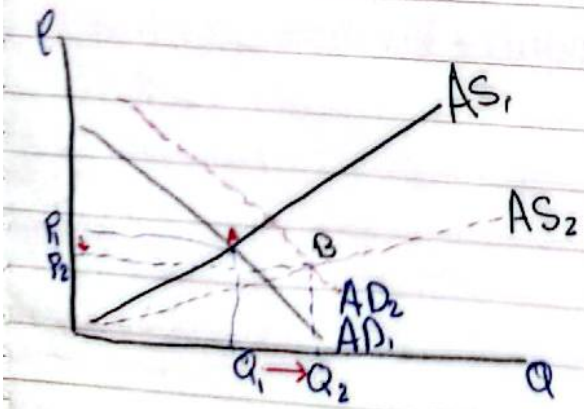
* increase in Q is greater than increase in P.
* happens at developed countries.

B-inelastic supply curve



* increase in Q is less than increas in P.
* happens at developing country

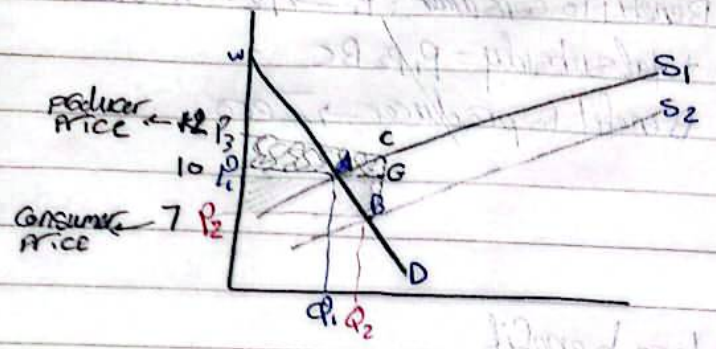
• Capital exp: $\uparrow G$



← with time.

* Supply curve will be more elastic
* positive effect is more than negative effect.

Ex) Competitive, Es > Ed, increasing MC, unit subsidy on production.



before subsidy:

Q → Q1 P → P1 Consumer surplus → AP1W

after subsidy

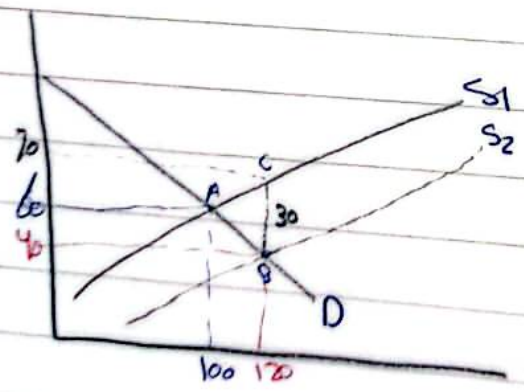
Q → Q1, P → P2, Consumer surplus → WBP2, Change in Consumer surplus → ABP2P1

unit subsidy → BC

total subsidy → BC * Q2 = P3CBP2

Consumer	Producer
10-7=3	12-10=2
BGP1P2	GCP3P1

Ex) Competitive ES > ED, mc ↑, unit subsidy



∴ Δ in consumer surplus = ABP1P2

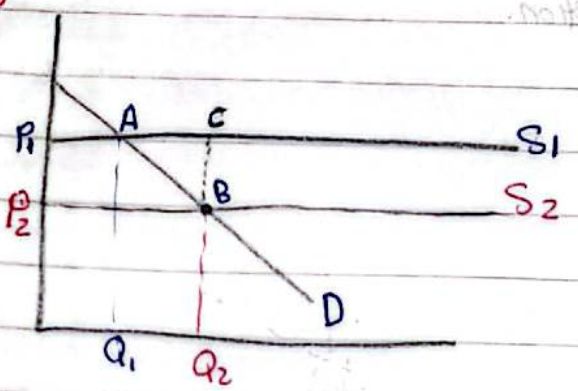
$\frac{1}{2}$ 2 base x hieght = $\frac{1}{2}(100+120) \times 20 = 2200$

- total subsidy = BC × Q2 = BCBP2 = 30 × 120 = 3600
- Benefit to Consumer = 60 - 40 = 20

  20 × 120 = 2400
- Benefit to producer = 70 - 60 = 10

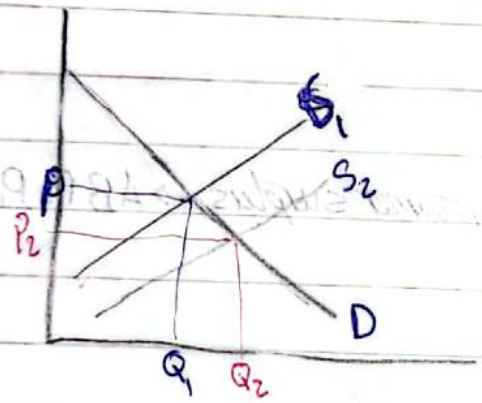
  10 × 120 = 1200

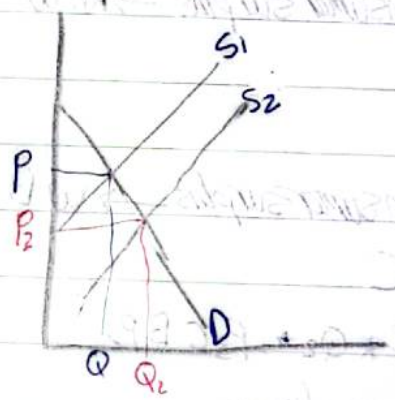
(2k)



Benefit to consumer : $P_1 \rightarrow P_2 \rightarrow P_1P_2BC$
total subsidy = P_1P_2BC
Benefit to producer → Zero

*more elasticity, less benefit.





*more ~~benefit~~ elasticity, less benefit

benefit received — earning/income

* $B = G - TE$

basic grant — reduction rate

- $G = 300$, $T = 25\%$, $B = ??$

① $E = 0$

$B = G - TE$

$= 300 - 0 = 300$

Total income after subsidy $= 0 + 300 = 300$

② $E = 500$

$B = G - TE$

$= 300 - 25\% \times 500 = 175$

total income $= 500 + 175 = 675$.

③ $E = 1000$

$B = G - TE$

$= 300 - 25\% \times 1000 = 50$

total income $= 1050$

④ $E = 1200$

$B = G - TE$

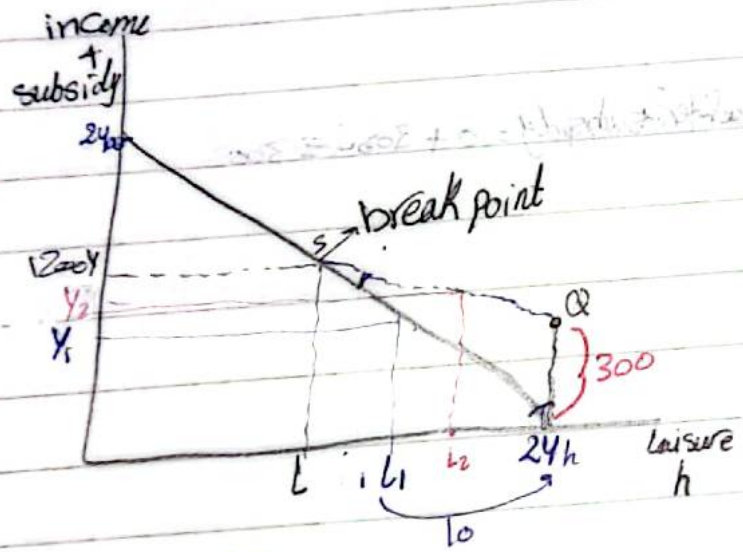
$300 - 25\% \times 1200 = 0$

total income $= 0 + 1200 = 1200$

* 24 h

Laisure	working	income working hour x wage (100)
24	0	0 x 100 = 0
12	12	12 x 100 = 1200
0	24	24 x 100 = 2400 → max

G = 300



New budget line = ASQ

(1) wage / hour = 200 , Leisure hour = 14, working hour = 10, G = 600, T = 20%

① earning before grant = 10 X 200 = 2000.

② total earned after grant = $\begin{cases} \text{income from work} = 2000 \\ \text{grant} = B = 600 - 2000 \times 20\% = 200 \end{cases}$ = 2000 + 200 = 2200

③ B = 0 when

600 - 20% X E = 0

$\frac{600}{20\%} = E$

E = 3000.

④ working hours = $\frac{3000}{200}$ = 15

Leisure hours = 9 h.

⑤ B = 600 - 25% X 2000 = 100.

(2) wage/hour = 100 , Leisure hour = 14 , working h = 10 , G = 500 , T = 25%

① 24 - 14 = 10. (L,T)

② Y_1 → Located on old BL.

Y_1 = 10 X 100 = 1000.

③ Y_2 → located on new BL < work + grant (benefit)

Y_2 = 1000 + [500 - 25% X 1000] = 1250.

④ break level $\Rightarrow$ S

⑤ $500 - 25\% \times E = 0$

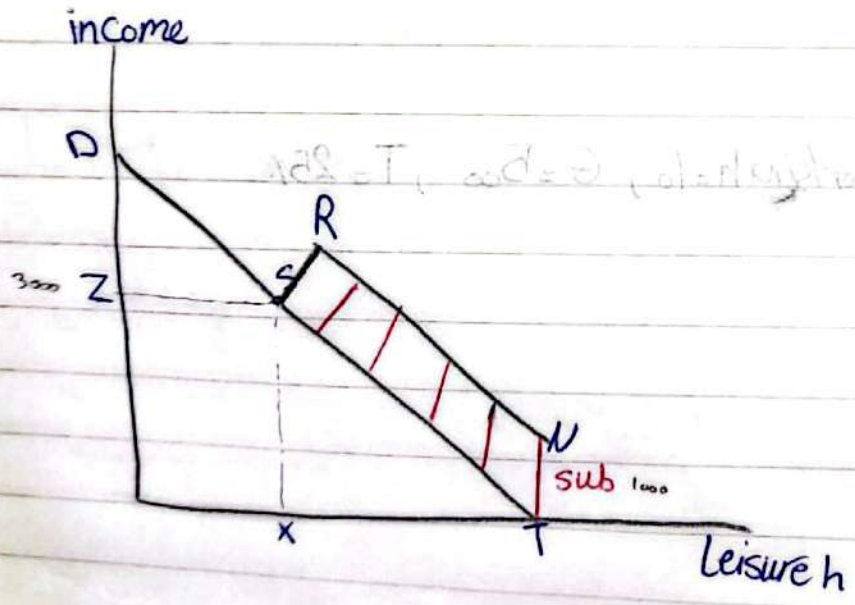
$E = \frac{500}{25\%} = 2000.$

⑥ work $h = \frac{2000}{100} = 20$

Leisure $h = 4.$

⑦ $B = 500 - 20\% \times 1000 = 300$

medicate survice / medicate insurance



new BL $\Rightarrow$ NRSD